THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 1 & 2: **Exploring Substances: Acidic, Basic, and Neutral, Electricity: Circuits and Their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** blue	bases change red litmus to blue
2	**(d)** neutral	no change in litmus
3	**(c)** citric acid	oranges, lemon
4	**(a)** stays yellow	no change in acids
5	**(a)** acid in the stomach	treated with antacids
6	**(b)** filament	heated by current
7	**(b)** switch	controls current
8	**(c)** conductors	metals
9	**(a)** Plastic is an insulator and protects from shock	safety
10	**(d)** positive terminal to negative terminal through the circuit	conventional direction

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	By adding lime, a base (2).	2
12	A substance whose smell changes in acid or base (1); onion or vanilla (1).	1+1
13	Two points called positive (+) and negative (−) (2).	2
14	Water conducts electricity so current may pass through the body (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Do not taste (1); wear gloves and goggles (1); handle with care (1).	1+1+1
16	More energy pushes more current (3).	3
17	Lemon has citric acid (1.5); soap is a base (1.5).	1½+1½
	OR: Acids react with marble and stone (1.5); cause erosion (1.5).	1½+1½
18	Wires, cooking vessels; plastic coating, handles (3).	3
	OR: Steel spoon and pencil lead (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Smell changes (1). (ii) Strips kept overnight and tested (2). (iii) Smell lost with base, stays with acid (2).	1+2+2
	OR: (i) Indicator behaviour (1). (ii) Acidic (1). (iii) Adding lime (1). (iv) Red cabbage (2).	1+1+1+2
20	(i) Continuous path (2). (ii) Current stops (2). (iii) Switch or fuse (1).	2+2+1
	OR: (i) Two or more cells (1). (ii) Positive to negative (1). (iii) Cancels voltage (1). (iv) Dimmer or off (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** basic substances	neutralisation protects water bodies
2	**(b)** Bases	typical properties
3	**(b)** sour	acids taste sour
4	**(b)** basic	base test
5	**(a)** acids formed from sugar	bacteria make acids; toothpaste is mildly basic
6	**(b)** Switch connected to mains	water conducts electricity
7	**(d)** battery	combination of cells
8	**(b)** a gap in the path	current cannot flow
9	**(d)** A rubber band as connecting wire	rubber is an insulator
10	**(b)** filament	glows white hot

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Neutral (1); sugar solution (1).	1+1
12	They are bases that neutralise excess stomach acid (2).	2
13	Cell and battery; mains supply (2).	2
14	It melts and breaks the circuit when current is too high (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Basic (1.5); baking soda solution (1.5).	1½+1½
16	Cell (energy), bulb (light), switch (control), wires/metal strips (path), reflector (focus) (3).	3
17	The base is neutralised slowly (1.5); the indicator changes colour at the neutral point (1.5).	1½+1½
	OR: Formic acid is injected (1.5); base neutralises it (1.5).	1½+1½
18	Insulation blocks contact so current cannot flow (3).	3
	OR: Skin and body fluids conduct and current harms (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Table of properties (2). (ii) Turns red with base (2). (iii) No, neutral (1).	2+2+1
	OR: (i) Formic acid (1). (ii) Baking soda paste (2). (iii) Neutralisation (2).	1+2+2
20	(i) Cell, bulb and wires with a gap (2). (ii) Cell, bulb, wires (1). (iii) Coin and key conduct; rubber does not (2).	2+1+2
	OR: (i) No wet hands, no bare wires (2). (ii) Prevent shocks and short circuits (2). (iii) Report and switch off (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 1 & 2: **Exploring Substances: Acidic, Basic, and Neutral, Electricity: Circuits and Their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** Lemon juice	citric acid gives the sour taste
2	**(b)** acetic acid	ethanoic acid
3	**(c)** formic acid	injected by the ant
4	**(c)** dark pink or magenta	natural indicator
5	**(a)** basic	they neutralise the excess acid
6	**(a)** circuit	closed loop
7	**(b)** a long line and a short line	long = positive
8	**(b)** Copper wire	metal
9	**(b)** copper	a good conductor
10	**(c)** alternating current (AC)	mains supply

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	To neutralise it and protect water bodies (2).	2
12	Dip filter paper in turmeric paste, dry and cut into strips (2).	2
13	It has polarity; the longer leg must go to the positive terminal (2).	2
14	Yes (1); graphite is a non-metal that conducts (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Shows the point where the solution becomes neutral (3).	3
16	Circle with cross, several cell symbols, break with lever (3).	3
17	Many plant materials change colour (1.5); turmeric and red cabbage (1.5).	1½+1½
	OR: A neutral solution does not taste sour (1); does not change the colour of litmus (1); an acidic solution tastes sour and turns blue litmus red (1).	1+1+1
18	No; rubber is an insulator (3).	3
	OR: Conductor core carries current; insulation covers it for safety (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Sour taste (2). (ii) Base reacts with acid to make bread soft (2). (iii) Both are useful (1).	2+2+1
	OR: (i) P and S acids; Q soap; R sugar (2). (ii) P and S (1). (iii) R (1). (iv) Q (1).	2+1+1+1
20	(i) Cells joined in series (2). (ii) Positive of one to negative of other (2). (iii) Excess current can fuse the bulb (1).	2+2+1
	OR: (i) Bare wires carry high current (1). (ii) AC (1). (iii) Insulators (1). (iv) Electricity department (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** They may be harmful	safety
2	**(a)** basic	turns red litmus blue
3	**(b)** a mild base such as an antacid	neutralises stomach acid
4	**(b)** changes colour with acids and bases	natural indicator
5	**(c)** neutral	neutralisation
6	**(b)** can conduct electricity	safety
7	**(b)** tungsten	high melting point
8	**(a)** Switch	ON/OFF
9	**(b)** current passes through wet skin	body conducts
10	**(d)** the metal base and the tip	contact points

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It turns dark pink/magenta (2).	2
12	Acidic: lemon, vinegar; basic: soap, baking soda; neutral: sugar, salt (2).	2
13	A complete path allowing current to flow (2).	2
14	It is a good conductor and ductile (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lime or slaked lime (1); it is a base and neutralises the soil acidity (2).	1+2
16	4.5 V (3).	3
17	Milk of magnesia (1); it is basic and neutralises acid (2).	1+2
	OR: Soap making and antacids (3).	3
18	Lemon juice contains dissolved substances; pure water has none (3).	3
	OR: Connect wire to bulb base and cell terminals and side; touch bulb base to the other terminal (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Bases remove grease; acids remove stains and rust (3). (ii) Soap/washing soda and vinegar on stains (2).	3+2
	OR: (i) Rain made acidic by gases from industries (1). (ii) Erodes marble, harms fish (2). (iii) Less burning of fossil fuels; use cleaner energy (2).	1+2+2
20	(i) Use a tester (3). (ii) Graphite, salt water (2).	3+2
	OR: (i) Standard symbols (2). (ii) Cell, switch, bulbs one after another (2). (iii) Universal and simple (1).	2+2+1

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